SAFETY DATA SHEET - HYDRO FLOW 195

1. IDENTIFICATION OF THE HAZARDOUS CHEMICAL AND SUPPLIER

Product Name: **HYDRO FLOW 195**

Product Description: Microbiocide

Supplier: HYDRO FLOW PTY. LTD.
61-63 Overseas Drive Noble Park, Victoria 3174 Australia
Telephone No.: 1300 793 121
Fax No.: 1300 852 143
ABN No.: 45 078 256 578

Emergency Tel.: 1300 793 121 (ALL HOURS)

2. HAZARDS IDENTIFICATION

ADG Code: Classified as Dangerous Goods according to the Australian Dangerous Goods Code (ADG Code) for transport by road and rail.

GHS Classification: Hazardous according to the criteria of the Globally Harmonised System of Classification and Labelling of Chemicals (GHS).

GHS Hazard Classification:
Corrosive to Metals - Category 1:H290
Skin Corrosion/Irritation - Category 1B: H314
Serious Eye Damage/Irritation - Category 1:H318
Acute Hazard to the aquatic environment - Category 1: H400
Chronic Hazard to the aquatic environment – Category 1: H410

LABEL ELEMENTS:

Hazard Pictogram:



Signal Word: Danger

Hazard Statement/s
AUH031 Contact with acids liberates toxic gas.
H290 May be corrosive to metals.
H314 Causes severe skin burns and eye damage.
H400 Very toxic to aquatic life.
H410 Very toxic to aquatic life with long lasting effects.

Precautionary Statement/s
P260 Do not breathe dust or mist. P234 Keep only in original packaging.
P273 Avoid release to the environment.
P280 Wear protective gloves/ protective clothing/ eye protection/ face protection. P264 Wash hands thoroughly after handling.

Response: P301+P330+P331 IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.
P303 + P361 + P353 IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower.
P363 Wash contaminated clothing before reuse.
P304 + P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P310 Immediately call a POISON CENTER or doctor/physician.
P321 Specific treatment is advised – see first aid instructions.
P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P390 Absorb spillage to prevent material damage.

Storage P403 + P233 Store in a well-ventilated place. Keep container tightly closed.

Disposal P501 Dispose of contents/container in accordance with local / regional / national /international regulations.

Poisons Schedule S5

3. COMPOSITION / INFORMATION ON INGREDIENTS

Component	CAS No.	Concentration	Classification
Sodium hypochlorite	7681-52-9	10-30%	Skin Corr. 1B: H314 Eye Damage 1: H318 Aquatic Acute 1: H400 Aquatic Chronic 1:H410
Sodium Hydroxide	1310-73-2	<1%	Skin Corr. 1A: H314
Water	7732-18-5	Balance	

4. FIRST AID MEASURES

Inhalation Remove victim from exposure to fresh air; avoid becoming a casualty. Remove contaminated clothing and loosen remaining clothing. Allow patient to assume most comfortable position and keep warm. Keep at rest until fully recovered. Seek medical advice if symptoms persist.

Ingestion Immediately rinse mouth out with water. Give plenty of water to drink. Do not induce vomiting. If vomiting occurs give further water. Seek immediate medical advice.

Skin Remove all contaminated clothing without delay. Wash skin gently and thoroughly with copious amounts of water and swab with polyethylene glycol 400. If swelling, blistering or redness occurs, seek medical attention. For skin burns, immediately flood burnt area with copious amounts of water and cover with a clean, dry dressing. Ensure contaminated clothing is washed before reuse or discarded. Seek medical attention.

Eye	Immediately flush eyes with plenty of water for at least 15 minutes, holding eyelids open. Seek immediate medical attention.
Symptoms/ effects from exposure	Treat symptomatically. Do not use acid antidotes in the treatment of sodium hypochlorite poisoning.
Medical treatment	None reasonably for seeable.
First Aid Facilities	Availability of eyewash and safety shower recommended.

5. FIRE FIGHTING MEASURES

Flash Point:	Not relevant
Flammability:	Non-flammable liquid
Hazchem Code:	2X
Suitable Extinguishing Media:	Water spray, fog, or mist, foam, carbon dioxide or dry chemical powder.
Specific Hazards:	Combustion or contact with strong acids may produce toxic chlorine gas.
Protective Equipment for firefighters:	Wear self-contained breathing apparatus and protective suit if risk of exposure to vapour or products of combustion exists.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions:	Personnel involved in the clean up should wear full protective clothing to avoid eye or skin contact. Eliminate all sources of ignition. Evacuate all unnecessary personnel. Spilt product may be slippery. Increase ventilation. Do not add acid.
Environmental Precautions	Avoid contaminating drains and waterways.
Cleaning up spills:	Contain to prevent run off into drains and waterways. Use non combustible absorbent (soil or sand or other inert material) to contain spill. Allow controlled access to drain accompanied by suitable neutralising agents such as sodium metabisulphite or sodium thiosulphate and a large excess of water. Collect and seal in properly labelled drums for disposal. Wash area down with excess water to remove residual material. For large spills notify local emergency services.

7. HANDLING AND STORAGE

Precautions for Safe Handling Product sensitive to mechanical impact and static discharge. Avoid contact with eyes, skin and clothing and repeated or prolonged skin contact. Prevent buildup of mists or vapours in the work atmosphere. Do not inhale vapour/fumes. When diluting the solution, add material to water.

Conditions for Safe Storage Store in a cool, dry, well-ventilated area. Keep containers tightly closed when not in use. Store away from incompatible materials including strong oxidizers, strong acids, metals, peroxides, reducing agents, metal salts and most metals. Protect from direct sunlight and store at ambient temperatures. Containers should always be vented to avoid buildup of pressure, and stored upright. Do not store in metal containers.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Exposure Limits: No value assigned for this specific material by Safe Work Australia. However, the following exposure standard applies to a decomposition product: Chlorine gas [CAS 7782-50-5] TWA = 3 mg/m³ (1ppm) peak limitation. Sodium hydroxide: Peak Limitation = 2 mg/m³.

Engineering Controls: Local and/or general exhaust is recommended. Keep containers closed when not in use.

Personal Protective Equipment:

Eyes: Wear chemical resistant safety glasses or full face shield.

Hands: Wear elbow length PVC, rubber or neoprene gloves.

Body: Wear waterproof apron, coveralls, trousers, long sleeved shirt, closed in shoes and/or safety footwear.

Respiratory: If inhalation risk from spray or mists exists wear a half face-piece filter respirator meeting the requirements of AS/NZ 1715 and 1716.

9. PHYSICAL AND CHEMICAL PROPERTIES

Form Liquid

Appearance Pale yellow-green clear liquid

Odour Slight chlorine

Boiling Point Ca 100°C (decomposes)

Freezing Point: Ca. -25°C

Solubility in Water miscible

Specific Gravity 1.2 at 20°C

pH Value 12-13

Vapour Pressure	2.3 kPa at 25°C
Relative Vapour Density	2.49; Air = 1
Volatile Component	No data
Flammability	Non flammable liquid

NOTE: The physical data presented above are typical values and should not be construed as a specification.

10. STABILITY AND REACTIVITY

Chemical Stability	The product is stable under normal conditions of use, storage and temperature.
Hazardous Reactions	When heated, or on contact with acids or acid vapours, produces toxic vapours of chlorides and chlorine gas. Reacts with nitrogen compounds to form unstable or explosive N-chloro compounds. Risk of explosion when in contact with methanol or benzyl cyanide. Violent reaction with hot formic acid.
Conditions to avoid	Avoid excessive heat, direct sunlight, moisture, freezing and static charges. Avoid contamination of solution, shock and friction.
Incompatible Materials	Avoid strong oxidizing agents, strong acids, metals such as aluminium, tin and zinc, peroxides, reducing agents, metal salts and sources of ignition. Avoid organic materials and foodstuffs. Reacts with acids forming toxic gases.
Hazardous Decomposition	When in a fire or in contact with strong acids, product may liberate toxic chlorine gas. When in contact with incompatible materials, reaction may produce flammable and highly explosive hydrogen gas.

11. TOXICOLOGICAL INFORMATION

Acute Oral Toxicity	LD50: (rat, oral): 5,300 mg/kg LD50: (rat, oral): 5,800 mg/kg (RTECS).
Ingestion	May be harmful if swallowed. Will cause severe irritation and corrosion (chemical burns) of the mucous membranes of the mouth, throat and gastrointestinal tract with nausea, vomiting, abdominal pain and inflammation. Systematic effects include fall of blood pressure, delirium and coma.
Inhalation	Inhalation of mists or vapours will result in respiratory irritation, cough, shortness of breath and possible harmful corrosive effects including lesions of the nasal septum, pulmonary oedema, pneumonitis and emphysema.

Skin	Corrosive to skin. Product may cause burns. Contact with skin will cause redness, itching, irritation, severe pain and chemical burns with resultant tissue destruction.
Eye	A severe eye irritant will cause stinging, blurring, tearing, severe pain. Causes serious eye damage. Corrosive to eyes, contact can cause corneal burns. Can result in permanent injury.
Skin sensitisation	Not known to be sensitizing.
Chronic effects	Repeated or prolonged skin contact may cause drying, cracking and irritation possibly resulting in development of dermatitis.
Carcinogenicity	No data available

12. ECOLOGICAL INFORMATION

Ecotoxicity	Forms corrosive mixtures with water even if diluted. Harmful effect due to pH shift. Aquatic Toxicity: 0.08 mg/L/96 hours. (fathead minnow pimephales promelas, LC50). 0.04 mg/L/48 hours (water flea daphnia magna, LC50). Toxicity to Bacteria: 100 mg/L/15 minutes (Photobacterium phosphoreum, EC50).
Persistence/ Degradability	No data.
Mobility	No data.
Environmental Fate	Very toxic to aquatic life. Avoid contaminating drains, waterways and sewers.
Bioaccumulation	No data.

13. DISPOSAL CONSIDERATIONS

Waste Disposal	Consult an EPA accredited waste disposal agent regarding recycling or disposal options.
Container Disposal	Rinse out container with water and dispose of in accordance with local regulations.

14. TRANSPORT INFORMATION

Road & Rail Transport	Dangerous Goods of Class 8 Corrosives are incompatible in a placard load with any of the following: - Class 1, Class 4.3, Class 5, Class 6, if the Class 6 dangerous goods are cyanides and the Class 8 dangerous goods are acids and Class 7.

U.N. Number	1791
Proper Shipping Name	HYPOCHLORITE SOLUTION
DG Class	8
Hazchem Code	2X
Packing Group	II
IERG Number	37

15. REGULATORY INFORMATION

Regulatory Information	Ingredients are listed on Australian Inventory of Chemical Substances (AICS).
Poisons Schedule	Product is listed as S5 in Australian Standard for Uniform Scheduling of Medicines and Poisons.

16. OTHER INFORMATION

Date of Preparation	24-01-2023 Version 4.0

End of Safety Data Sheet